

Diagnosing Invalid Simplification of Sums of Radicals

The one rule that does not exist:

- $\sqrt{a} + \sqrt{b}$ is *not* $\sqrt{a+b}$. Roots distribute over multiplication, never over addition.
- Two radical terms combine only when their radicands are identical *after* each one is simplified: $\sqrt{8} + \sqrt{18}$ looks unlike, but both hide $\sqrt{2}$.
- When they do combine, you add the coefficients and the radicand is untouched: $2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$, not $5\sqrt{4}$.

Give exact answers in simplest radical form. Where a decimal is asked for, round to two decimal places — a decimal is how you *test* a claim, not how you state the answer.

1. Write $\sqrt{72}$ in simplest radical form.

Answer: _____

2. Simplify $5\sqrt{3} + 2\sqrt{3} - \sqrt{3}$.

Answer: _____

3. Simplify $\sqrt{8} + \sqrt{18}$ in simplest radical form. Then give its decimal value to two places, and say in one line why the answer is not $\sqrt{26}$.

Answer: _____

4. Find and fix the mistake. A student writes

$$\sqrt{9} + \sqrt{16} = \sqrt{25} = 5$$

Compute the left side correctly, state which of the two values is greater, and name the rule the student invented.

Answer: _____

5. Find and fix the mistake. A student simplifies $4\sqrt{5} + 3\sqrt{2}$ to $7\sqrt{7}$. Explain what was added that should not have been, decide whether $4\sqrt{5} + 3\sqrt{2}$ can be simplified at all, and give its decimal value to two places.

Answer: _____

6. Simplify $\sqrt{12} + \sqrt{27} - \sqrt{3}$ in simplest radical form, and give its decimal value to two places. Show the simplified form of each radical before you combine anything.

Answer: _____

7. Find and fix the mistake. A student claims $\sqrt{2} + \sqrt{8} = \sqrt{10}$. Give the decimal value of $\sqrt{2} + \sqrt{8}$ to two places, write the correct comparison symbol between $\sqrt{2} + \sqrt{8}$ and $\sqrt{10}$, and write $\sqrt{2} + \sqrt{8}$ in simplest radical form.

Answer: _____

8. Explain. Write a short paragraph for a classmate who believes that square roots may be added the way fractions with the same denominator are. Say why $\sqrt{a} + \sqrt{b}$ is not $\sqrt{a+b}$, state exactly when two radical terms may be combined, and give one numerical counterexample with the arithmetic shown.